EXPERIMENT 8

User Authentication using JWT and Form-Based Login

AIM

To implement user registration and login using password hashing and JSON Web Tokens (JWT).

ALGORITHM

1. Accept user registration details from the frontend.
2. Hash the password using bcrypt before storing it.
3. Compare the entered password during login.
4. Generate a JWT after successful authentication.
5. Verify the JWT in middleware before allowing protected access.

PROGRAM

```
// Install: npm install express bcryptjs jsonwebtoken
// server.js (demo using in-memory storage)
const express=require('express'),bcrypt=require('bcryptjs');
const jwt=require('jsonwebtoken');
const app=express(); app.use(express.json());
const users=[], SECRET='change_this_secret';

app.post('/register',async(req,res)=>{
 const hash=await bcrypt.hash(req.body.password,10);
 users.push({username:req.body.username,hash});
 res.json({message:'Registered'});
});
app.post('/login',async(req,res)=>{
 const u=users.find(x=>x.username===req.body.username);
 if(!u || !(await bcrypt.compare(req.body.password,u.hash)))
  return res.status(401).json({message:'Invalid login'});
 res.json({token:jwt.sign({username:u.username},SECRET,{expiresIn:'1h'})});
});
function auth(req,res,next){
 try{req.user=jwt.verify(req.headers.authorization.split(' ')[1],SECRET);next();}
 catch{res.sendStatus(401);}
}
app.get('/profile',auth,(req,res)=>res.json(req.user));
app.listen(5000);
```

RESULT

Thus, form-based user authentication using bcrypt password hashing and JWT verification was implemented successfully.

VIVA QUESTIONS

1. What is JWT?
2. What is the difference between authentication and authorization?
3. Why are passwords hashed?
4. What is the purpose of JWT middleware?
5. What is the Authorization header?